

Minimum Input Complexity for Separating Qubit POVMs from Projective Measurements

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Abstract

Earlier work showed that generalized qubit measurements can outperform projective qubit measurements when one party has three Bell inputs and the other has two. We prove the matching lower bound: two inputs per party never suffice. For every choice of finite, input-dependent output alphabets, every bipartite behavior generated by qubit positive-operator-valued measurements belongs to the shared-randomness convex hull of behaviors generated by qubit projection-valued measurements. The statement permits arbitrary two-qubit mixed states, zero projectors, deterministic relabeling, and stochastic postprocessing, while forbidding dimension-increasing dilation. We also give an explicit rational Bell functional with three inputs for one party and two for the other whose value under an exact qubit POVM strategy strictly exceeds a global analytic upper bound for every qubit projective strategy. Thus 3×2 is the minimum input architecture, up to exchanging the parties. The equality proof reduces every possible two-input separator to a binary- ternary residual architecture, models that architecture by a Lorentz incidence manifold, obtains strictly positive multipliers from finite POVM duality, and eliminates all ranks of the metric differential by second-order geometry or explicit projective simulation.

1 Introduction

A positive-operator-valued measurement (POVM) is more general than a projection-valued measurement (PVM) on a fixed Hilbert space. Naimark dilation represents a POVM projectively only after adding degrees of freedom, so it does not preserve a fixed local dimension. Bell experiments therefore offer an operational question: how many measurement inputs are needed before generalized measurements on local qubits generate correlations unavailable to projective qubit measurements? This belongs to the broader study of Bell nonlocality and dimension-constrained quantum behavior [1, 2].

Earlier work established that nonprojective qubit measurements can be useful in Bell scenarios with three inputs on one side and two on the other [3, 4], and related device-independent constructions certify nonprojective qubit measurements with larger input architectures [5]. This does not settle the two-input behavior-set question. It also differs from operator-level projective simulability of individual POVMs [6, 7]: a Bell behavior may admit a projective realization using a different state even when the original local POVM is not itself a mixture of PVMs.

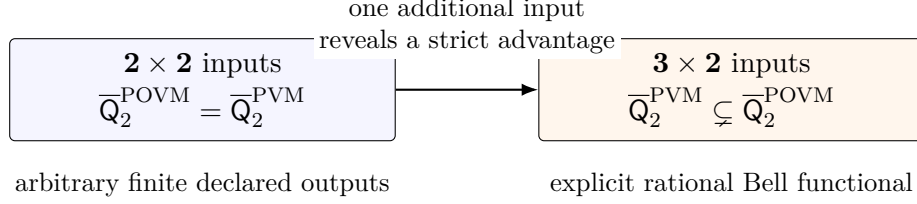


Figure 1: The minimum input threshold at fixed local dimension two. Equality means equality after shared-randomness convexification for each fixed finite declared-output architecture.

Strong reductions are known when every measurement is dichotomic [8, 9]; genuine ternary effects are precisely what those arguments do not address. They also make exhaustive PVM conventions essential. A published ternary-measurement claim was later corrected after a qubit PVM with a zero effect was found to attain the same value [10, 11]. We therefore retain zero projectors, deterministic inputs, every support pattern, and stochastic postprocessing throughout.

Contributions

We work throughout with local Hilbert-space dimension at most two and with shared-randomness convexification. Our results are:

1. An explicit rational 3×2 Bell functional has an exact algebraic qubit-POVM strategy above a global analytic upper bound for all qubit-PVM strategies (theorem 3.1).
2. If one party has binary outcomes on both of its two inputs, every qubit POVM behavior is a common convex mixture of complete qubit-PVM behaviors (theorem 4.1).
3. Any remaining strict two-input separator reduces to a pure entangled strategy with one binary PVM and one ternary rank-one POVM per party (theorem 4.3).
4. The residual architecture cannot strictly exceed the PVM support function (theorem 9.3). The proof treats every smooth and singular rank of an exact Lorentz incidence model.
5. Consequently 3×2 is the minimum input architecture, up to exchanging the parties (corollary 9.4).

The equality is not an operator identity between POVMs and PVMs, nor does it assert simulation on the same state. It is an equality of convex behavior sets. The distinction is central.

Proof roadmap

The exact separation combines a CHSH locking term with a three-state discrimination task. For the equality theorem, convex separation first turns any counterexample into a global Bell maximizer. A Lorentz-cone circuit argument handles every case in which one party is binary. A common-span filtering perturbation and extremal-POVM dimension bound reduce all other counterexamples to one binary and one ternary measurement per party.

In the residual architecture, the five local effects form a unique binary-versus-ternary circuit. Lorentz coordinates describe the complete strategy by a 4×4 probability block P and a four-parameter metric g , subject to five quadratic null equations. Finite POVM duality makes every determinant multiplier strictly positive at a hypothetical strict separator. The constrained second

variation then has ambient inertia (4, 12). If the metric differential has rank at least two, dimension alone forces an uphill direction. Rank one contradicts positivity through an exact projective-fiber theorem, while rank zero has an explicit deterministic decomposition.

2 Model and convexified behavior sets

Let $X = Y = \{0, 1\}$ unless stated otherwise. Each input has a finite declared output alphabet A_x or B_y , which may depend on the input.

Definition 2.1 (Fixed-qubit behavior). A single fixed-qubit POVM strategy consists of Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$ of dimensions at most two, a density operator ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$, and effects

$$M_a^x \geq 0, \quad \sum_{a \in A_x} M_a^x = \mathbb{I}_A, \quad N_b^y \geq 0, \quad \sum_{b \in B_y} N_b^y = \mathbb{I}_B.$$

It produces

$$p(a, b|x, y) = \text{Tr}[\rho(M_a^x \otimes N_b^y)].$$

The resulting set is denoted $\mathbf{Q}_2^{\text{POVM}}(\mathbf{A}, \mathbf{B})$.

A PVM strategy requires all effects in each measurement to be mutually orthogonal projections. Zero projections are permitted. Therefore a qubit PVM with arbitrarily many declared labels has at most two nonzero effects. No ancillary system or Naimark dilation may increase a local dimension above two.

We denote the shared-randomness hulls by

$$\overline{\mathbf{Q}}_2^{\text{POVM}}(\mathbf{A}, \mathbf{B}) := \text{conv } \mathbf{Q}_2^{\text{POVM}}(\mathbf{A}, \mathbf{B}), \quad \overline{\mathbf{Q}}_2^{\text{PVM}}(\mathbf{A}, \mathbf{B}) := \text{conv } \mathbf{Q}_2^{\text{PVM}}(\mathbf{A}, \mathbf{B}).$$

Local stochastic postprocessing is included on the PVM side.

Lemma 2.2 (Convexification and postprocessing). *For a linear Bell functional, shared randomness and stochastic postprocessing do not increase the optimum beyond one deterministic projective component.*

Proof. Linearity gives

$$\mathcal{F}\left(\sum_{\lambda} q_{\lambda} p_{\lambda}\right) = \sum_{\lambda} q_{\lambda} \mathcal{F}(p_{\lambda}) \leq \max_{\lambda} \mathcal{F}(p_{\lambda}).$$

A finite stochastic map is a convex combination of deterministic maps. Absorb all deterministic choices into the shared variable. A deterministic map merges mutually orthogonal projectors assigned to the same displayed label and inserts zero projectors for unused labels, hence remains a declared- output PVM. $\square$

For fixed finite alphabets, all strategy spaces above are compact. Indeed, density matrices and tuples of effects form closed bounded sets in finite-dimensional matrix spaces, and the Born map is continuous. The convex hull of a compact subset of a finite-dimensional affine space is compact: affine dependence reduces every convex combination to at most $m + 1$ points in affine dimension m .

Lemma 2.3 (Nearest-point separation). *Let $\mathcal{P}$ be a compact convex subset of a finite-dimensional Euclidean space. If $q \notin \mathcal{P}$ and p_0 minimizes $\|q - p\|$ over $p \in \mathcal{P}$, then $c = q - p_0$ satisfies*

$$c \cdot q > \sup_{p \in \mathcal{P}} c \cdot p.$$

Proof. For $p \in \mathcal{P}$, differentiate $\|q - p_0 - t(p - p_0)\|^2$ at $t = 0^+$. Minimality gives $(q - p_0) \cdot (p - p_0) \leq 0$, and strictness follows from $q \neq p_0$. $\square$

Remark 2.4 (Notation for arbitrary outputs). The shorthand $\overline{\mathbf{Q}}_2^{\text{POVM}}(2, 2) = \overline{\mathbf{Q}}_2^{\text{PVM}}(2, 2)$ means an equality schema: the equality holds separately for every choice of finite $(A_0, A_1; B_0, B_1)$. It is not an identification of behavior vectors living in different ambient spaces.

3 An exact rational 3×2 separation

For this section take Alice inputs $x = 0, 1, 2$, Bob inputs $y = 0, 1$, Alice labels $a = 0, 1, 2$, and Bob labels $b = 0, 1$. Define signs

$$\alpha_0 = \alpha_2 = 1, \quad \alpha_1 = -1, \quad \beta_0 = 1, \quad \beta_1 = -1$$

and correlators

$$E_{xy} = \sum_{a,b} \alpha_a \beta_b p(a, b|x, y).$$

Set

$$\begin{aligned} \mathcal{B}(p) = & 10(E_{00} + E_{01} + E_{10} - E_{11}) \\ & + \frac{3}{5}p(0, 0|2, 0) + \frac{3}{5}p(1, 1|2, 0) + \frac{4}{5}p(2, 0|2, 1). \end{aligned} \tag{1}$$

For this declared-output architecture, write

$$\beta_{\text{POVM}}(\mathcal{B}) := \sup_{p \in \overline{\mathbf{Q}}_2^{\text{POVM}}} \mathcal{B}(p), \quad \beta_{\text{PVM}}(\mathcal{B}) := \sup_{p \in \overline{\mathbf{Q}}_2^{\text{PVM}}} \mathcal{B}(p).$$

Theorem 3.1 (Exact fixed-qubit separation). *For the rational functional (1),*

$$\beta_{\text{POVM}}(\mathcal{B}) \geq L_1 := \frac{16 + 8\sqrt{7813}}{25} > U := 20\sqrt{2} + \frac{3}{5} + \frac{4 + 3\sqrt{2}}{250} \geq \beta_{\text{PVM}}(\mathcal{B}).$$

The certified gap is

$$L_1 - U = \frac{6 + 80\sqrt{7813} - 5003\sqrt{2}}{250} > 0.$$

These are an attained POVM lower bound and a global PVM upper bound; neither is asserted to be the corresponding exact optimum.

3.1 The simple exact strategy

Let X, Z be Pauli matrices and

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}.$$

Bob uses $B_0 = Z$, $B_1 = X$. Alice uses

$$A_0 = (Z + X)/\sqrt{2}, \quad A_1 = (Z - X)/\sqrt{2}$$

for her first two binary measurements, with a zero third label. Her final measurement is

$$\begin{aligned} M_0 &= (17\mathbb{I} - 8X + 15Z)/50, \\ M_1 &= (17\mathbb{I} - 8X - 15Z)/50, \\ M_2 &= (16\mathbb{I} + 16X)/50. \end{aligned}$$

The rank-one factorizations

$$M_0 = \frac{1}{25} \begin{pmatrix} 4 \\ -1 \end{pmatrix} (4, -1), \quad M_1 = \frac{1}{25} \begin{pmatrix} 1 \\ -4 \end{pmatrix} (1, -4), \quad M_2 = \frac{8}{25} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1, 1)$$

prove positivity and normalization. Their nonzero eigenvalues are $17/25, 17/25, 16/25$, so the measurement is not a PVM.

The identity

$$\langle \Phi^+ | R \otimes S | \Phi^+ \rangle = \frac{1}{2} \text{Tr}(R^T S)$$

gives CHSH value $2\sqrt{2}$ and

$$p(0, 0|2, 0) = p(1, 1|2, 0) = p(2, 0|2, 1) = \frac{8}{25}.$$

Thus the simpler value

$$L_0 = 20\sqrt{2} + \frac{16}{25}$$

already exceeds U , since

$$L_0 - U = \frac{3(2 - \sqrt{2})}{250} > 0.$$

At the ideal point the auxiliary term is a state-discrimination problem with

$$W_0 = \frac{3}{20}(\mathbb{I} + Z), \quad W_1 = \frac{3}{20}(\mathbb{I} - Z), \quad W_2 = \frac{1}{5}(\mathbb{I} + X).$$

The dual operator

$$\Gamma = \frac{8}{25}\mathbb{I} + \frac{2}{25}X$$

satisfies $\Gamma \geq W_i$ and $(\Gamma - W_i)M_i = 0$, so the POVM value is $16/25$. Every one- or two-label qubit PVM support scores at most $3/5$. The ideal discrimination advantage is therefore $1/25$.

3.2 Global projective bound

Write $\mathcal{B} = 10S + T$. For a PVM strategy, the displayed CHSH observables are Hermitian unitaries, including possible values $\pm\mathbb{I}$. If one is degenerate, the CHSH identity

$$C_{\text{CHSH}}^2 = 4\mathbb{I} - [A_0, A_1] \otimes [B_0, B_1]$$

gives $S \leq 2$, while $T \leq 7/5$, which is below U .

Otherwise optimize the state to a pure state and put it in Schmidt form

$$|\psi_\eta\rangle = \sqrt{\frac{1+\eta}{2}}|00\rangle + \sqrt{\frac{1-\eta}{2}}|11\rangle, \quad 0 \leq \eta \leq 1.$$

Its correlation matrix is

$$R = \text{diag}(\sqrt{1-\eta^2}, -\sqrt{1-\eta^2}, 1).$$

If $u = |\mathbf{b}_0 \cdot \mathbf{b}_1|$ and $\delta = 2\sqrt{2} - S$, optimizing Alice's directions and applying Cauchy–Schwarz gives

$$S \leq 2\sqrt{2} - \frac{\eta^2}{\sqrt{2}}, \quad S \leq 2\sqrt{2} - \frac{u^2}{2\sqrt{2}}, \quad (2)$$

hence

$$\eta \leq 2^{1/4}\sqrt{\delta}, \quad u \leq 2^{3/4}\sqrt{\delta}.$$

A ternary qubit PVM has one of the six labeled rank patterns obtained by permuting $(2, 0, 0)$ or $(1, 1, 0)$. For every singleton support and the pair $\{0, 1\}$, $T \leq 3/5 + 2(\eta + u)/5$. A pair involving label 2 reduces to binary discrimination between

$$\Pi(\mathbf{v}) := \frac{1}{2}(\mathbb{I} + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad A = \frac{3}{5}D\Pi(\mathbf{n})D, \quad B = \frac{4}{5}D\Pi(\mathbf{m})D, \quad D = \text{diag}\left(\sqrt{\frac{1+\eta}{2}}, \sqrt{\frac{1-\eta}{2}}\right).$$

With $x = n_z$, $y = m_z$, and $r = \mathbf{n} \cdot \mathbf{m}$,

$$20F_\eta = 7 + \eta(3x + 4y) + \sqrt{[-1 + \eta(3x - 4y)]^2 + 24(1 - \eta^2)(1 - r)}.$$

The exact comparison

$$F_\eta \leq F_0(r) + \frac{2\eta}{5} \leq \frac{3}{5} + \frac{2}{5}(\eta + u)$$

is proved by the manifestly nonnegative polynomial certificate in appendix A. Consequently

$$T \leq \frac{3}{5} + \frac{2}{5}(\eta + u) \leq \frac{3}{5} + C\sqrt{\delta}, \quad C = \frac{2}{5}(2^{1/4} + 2^{3/4}),$$

where $C^2 = (16 + 12\sqrt{2})/25$. Completing the square,

$$\mathcal{B} \leq 20\sqrt{2} + \frac{3}{5} + C\sqrt{\delta} - 10\delta \leq 20\sqrt{2} + \frac{3}{5} + \frac{C^2}{40} = U.$$

This covers every state, support pattern, direction, relabeling, and degeneracy. Lemma 2.2 covers shared randomness and postprocessing. The stronger algebraic strategy attaining L_1 and exact gap comparison are given in appendix B.

4 Two-input reductions

4.1 One binary party

Theorem 4.1 (One-binary-party simulation). *Suppose one party in a two-input bipartite qubit scenario has binary declared outputs on both inputs. The other party may have arbitrary finite outputs. Then every qubit-POVM behavior belongs to the convex hull of fixed-qubit PVM behaviors. The statement is symmetric under exchanging the parties.*

Proof. Decompose each binary qubit POVM into PVMs. If E has eigenvalues $0 \leq \mu \leq \lambda \leq 1$ and P is the spectral projector for λ , then

$$\{E, \mathbb{I} - E\} = (\lambda - \mu)\{P, \mathbb{I} - P\} + \mu\{\mathbb{I}, 0\} + (1 - \lambda)\{0, \mathbb{I}\}.$$

Taking the product of the decompositions for both inputs reduces, in each shared-random component, Bob to fixed binary PVMs with observables B_0, B_1 .

For Alice outcome $a|x$, let $\sigma_{a|x}$ be Bob's subnormalized steered state and define

$$\Phi_B(\sigma) = (\text{Tr } \sigma, \text{Tr}(\sigma B_0), \text{Tr}(\sigma B_1)) \in \mathbb{R}^3.$$

These data determine all joint probabilities:

$$p(a, b|x, y) = \frac{1}{2} \left[(\Phi_B \sigma_{a|x})_0 + (-1)^b (\Phi_B \sigma_{a|x})_{y+1} \right].$$

No-signaling gives equality of the sums of the two compressed ensembles.

If $\mathbb{I}, B_0, B_1$ are linearly independent, the image of the positive qubit cone is a three-dimensional Lorentz cone: the cone over the elliptical projection of the Bloch ball onto the span of the two Bloch axes. Parallel, antiparallel, or scalar observables give a two-dimensional simplicial cone or a local deterministic degeneration; these cases are included in appendix C.

Decompose every compressed positive vector into extreme rays, retaining its original output label. Repeatedly extract support-minimal positive subrelations. In real dimension at most three, such a circuit has at most four distinct signed rays. A one-versus-many relation would express an extreme ray as a positive sum of different cone rays and is impossible. Collinear one-versus-one relations are zero-padded binary components. Every nontrivial circuit is therefore

$$r_0 + r_1 = s_0 + s_1.$$

Every extreme compressed ray has a unique positive rank-one lift inside $\text{span}_{\mathbb{R}}\{\mathbb{I}, B_0, B_1\}$. Thus the circuit lifts to

$$R_0 + R_1 = S_0 + S_1 = \Omega$$

between rank-one positive operators. If Ω is invertible, the two rank-one decompositions become two orthonormal bases after applying $\Omega^{-1/2}$. The canonical purification

$$|\Psi_{\Omega}\rangle = (\mathbb{I} \otimes \sqrt{\Omega/\text{Tr } \Omega})(|00\rangle + |11\rangle)$$

therefore realizes both ensembles by two Alice PVMs. Rank-one Ω , zero terms, and repeated rays require a little bookkeeping. If $\Omega = |\omega\rangle\langle\omega|$ has rank one, every nonzero summand is collinear with Ω , so

$$R_i = \rho_i \Omega, \quad S_i = \sigma_i \Omega, \quad \rho_0 + \rho_1 = \sigma_0 + \sigma_1 = 1.$$

A product purification with Bob state proportional to Ω , together with independently chosen Alice PVM bases, realizes the two arbitrary binary weight vectors (ρ_0, ρ_1) and (σ_0, σ_1) . Zero terms receive zero projectors. Collinear terms are combined only within their retained output labels; if the same geometric ray occurs under different labels, classical splitting restores those labels branchwise. Thus no outcome probability is lost when repeated rays are removed.

Give this circuit strategy shared weight $\text{Tr } \Omega$. Circuit elimination preserves total trace, so the weights sum to one. The retained output labels and fixed Bob PVMs reconstruct every original compressed vector and hence every joint probability. This is one common decomposition of the complete behavior, not a separate decomposition for each input. $\square$

4.2 Extreme residual architecture

Lemma 4.2 (Common-span filtering). *Let Alice's two POVMs have a common Hermitian combination*

$$H = \sum_a s_{a|0} M_{a|0} = \sum_a s_{a|1} M_{a|1}.$$

For sufficiently small $\varepsilon > 0$, define

$$S_{\pm} = \mathbb{I} \pm \varepsilon H, \quad r_{a|x}^{\pm} = 1 \pm \varepsilon s_{a|x},$$

$$M_{a|x}^\pm = r_{a|x}^\pm S_\pm^{-1/2} M_{a|x} S_\pm^{-1/2}.$$

With

$$q_\pm = \text{Tr}[(S_\pm \otimes \mathbb{I})\rho], \quad \rho_\pm = \frac{(S_\pm^{1/2} \otimes \mathbb{I})\rho(S_\pm^{1/2} \otimes \mathbb{I})}{q_\pm},$$

the original complete behavior satisfies

$$p = \frac{q_+}{2} p_+ + \frac{q_-}{2} p_-.$$

Proof. The effects normalize because $\sum_a r_{a|x}^\pm M_{a|x} = S_\pm$, and $q_+ + q_- = 2$. Direct cyclicity of trace gives

$$q_\pm p_\pm(a, b|x, y) = r_{a|x}^\pm p(a, b|x, y).$$

Averaging the signs proves the identity for all inputs and outputs. $\square$

Theorem 4.3 (Residual-architecture reduction). *If a Bell functional with two inputs per party has a strict qubit-POVM advantage over $\overline{\mathcal{Q}}_2^{\text{PVM}}$, a global maximizing component may be chosen with:*

- (i) a pure full-Schmidt-rank two-qubit state;
- (ii) one nondegenerate binary PVM on each party;
- (iii) one extremal ternary rank-one POVM on each party;
- (iv) local real measurement spans intersecting only in $\mathbb{RI}$.

In particular, four-outcome extremal POVMs are unnecessary.

Proof. Choose an extreme point of the exposed maximizing face. Compactness and extremality let us select one unrandomized realization. Decompose its state into pure states while holding the measurements fixed; every positive-weight component must produce the same extreme behavior. A product state is local, so select a pure entangled state, whose reduced states are full rank.

Likewise decompose each local POVM into extremal POVMs, one measurement at a time. Extremality permits an extremal component that still realizes the same behavior. Temporarily delete zero effects; they contribute no probabilities and their declared output labels can be reinserted at the end. Full-rank reduced states then make every remaining outcome marginal strictly positive. For one party set

$$V_x = \text{span}_{\mathbb{R}}\{M_{a|x}\} \subset \text{Herm}(2).$$

If $V_0 \cap V_1$ contained a nonscalar H , lemma 4.2 would split the behavior. Equality of both branch behaviors would imply, using strictly positive outcome marginals, $r_{a|x}^\pm = q_\pm$ for every nonzero outcome, making all coefficients equal and H scalar. Thus

$$V_0 \cap V_1 = \mathbb{RI}.$$

Since $\dim_{\mathbb{R}} \text{Herm}(2) = 4$, one of V_0, V_1 has dimension at most two. It is commuting and is a stochastic postprocessing of a binary spectral PVM. Behavior extremality selects a deterministic postprocessing, so this input may be taken as a PVM. Retain an extreme deterministic component realizing the same behavior and apply lemma 4.2 once more to that realization; its two final measurement spans therefore still intersect only in $\mathbb{RI}$. The PVM is nondegenerate, because a party with only one nontrivial input produces a local behavior.

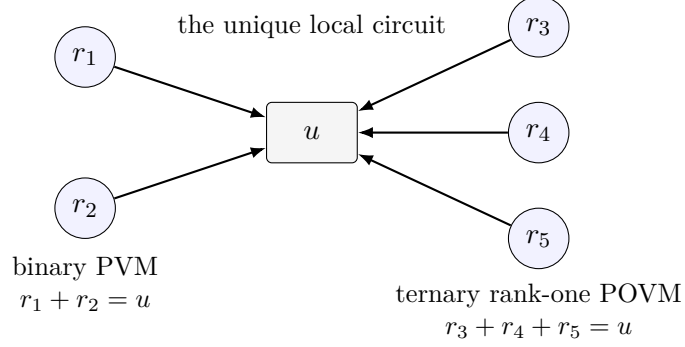


Figure 2: The local coefficient circuit after choosing four effects as an operator basis. The physical effects are future-null rays in a Lorentz metric.

The other span has dimension at most three. Its nonzero effects are linearly independent; otherwise lemma 4.2 with $H = 0$ splits the extreme behavior. For an extremal qubit POVM, support-preserving perturbations give

$$\sum_a \text{rank}(M_a)^2 \leq 4.$$

The standard extremality criterion [12, 13] is reproved in appendix D for completeness. A genuine nonprojective measurement in a three-dimensional span therefore has exactly three nonzero rank-one effects. If both spans on one party were at most two, theorem 4.1 would make the full behavior PVM-simulable. Apply the argument to both parties, then reinsert any deleted zero-output labels. $\square$

5 Lorentz incidence model

Write every Hermitian 2×2 matrix as

$$X = x_0 \mathbb{I} + x_1 \sigma_x + x_2 \sigma_y + x_3 \sigma_z$$

and set $J = \text{diag}(1, -1, -1, -1)$. Then

$$\det X = x^T J x.$$

Nonzero positive rank-one effects are precisely future-null vectors: $x_0 > 0$, $x^T J x = 0$.

In the residual architecture label one party's two binary effects by E_1, E_2 and ternary effects by E_3, E_4, E_5 . Set

$$r_1 = e_1, \quad r_2 = e_2, \quad r_3 = e_3, \quad r_4 = e_4, \quad r_5 = (1, 1, -1, -1)^T, \quad u = (1, 1, 0, 0)^T.$$

Their unique coefficient circuit is

$$r_1 + r_2 = r_3 + r_4 + r_5 = u.$$

The first four physical effects form a real basis of $\text{Herm}(2)$. Let $\mathbf{E}$ contain their Lorentz vectors as columns and define $g = \mathbf{E}^T J \mathbf{E}$. Nullness and normalization force

$$g = \begin{pmatrix} 0 & \frac{1}{2} & a & b \\ \frac{1}{2} & 0 & c & d \\ a & c & 0 & e \\ b & d & e & 0 \end{pmatrix}, \quad e = a + b + c + d - \frac{1}{2}. \quad (3)$$

On the genuine residual stratum all distinct-ray products are positive; in particular $a, b, c, d, e > 0$ and

$$a + b, c + d, a + c, b + d < \frac{1}{2}.$$

Choose the first four effects on both parties as bases and let P be their 4×4 joint-probability block. The complete table is

$$p_{ij} = r_i^T P r_j, \quad u^T P u = 1.$$

For a pure full-Schmidt-rank state $|\psi_C\rangle = \sum C_{mn}|mn\rangle$, steering is $\Theta_C(N) = C N^T C^\dagger$. Its Lorentz matrix L_C satisfies

$$L_C^T J L_C = |\det C|^2 J.$$

If h is Bob's local metric, then

$$P^T g^{-1} P = 4 |\det C|^2 h.$$

Thus P is invertible. With $Q = g^{-1}$, define

$$F_j(P, g) := (P r_j)^T Q (P r_j) = 0, \quad j = 1, \dots, 5. \quad (4)$$

6 Physical exactness and smoothness

Theorem 6.1 (Local physical completeness). *Near every genuine residual point, every nearby (P, g) satisfying (3), the five incidence equations (4), normalization, Lorentz signature, and the strict time-orientation inequalities is realized by a genuine residual two-qubit strategy. Every incidence tangent therefore integrates to a two-sided physical curve.*

Proof. Because g has signature $(1, 3)$, $u^T g u = 1$, and the restriction to $u^\perp$ is negative definite, Lorentz Gram-Schmidt gives a smoothly varying E with

$$E^T J E = g, \quad E u = (1, 0, 0, 0)^T.$$

Each r_i is g -null and has positive product with u ; hence $E r_i$ is future null and represents a positive rank-one Alice effect. The circuit makes both measurements sum to the identity.

Put $x_j = P r_j$ and

$$s_j = \frac{1}{2} E^{-T} x_j.$$

The incidence equations give $s_j^T J s_j = 0$, while $2(s_j)_0 = u^T x_j$ is the strictly positive Bob marginal. Thus s_j represents a future-null positive rank-one steered operator S_j . The circuit yields

$$S_1 + S_2 = S_3 + S_4 + S_5 =: \rho_A.$$

Normalization gives $\text{Tr } \rho_A = 1$, and full rank is open near the original point. Let $C = \rho_A^{1/2}$ and set

$$N_j^T = C^{-1} S_j C^{-\dagger}.$$

The N_j are positive rank-one effects and normalize on both inputs. Moreover

$$\text{Tr}(E_i S_j) = r_i^T P r_j,$$

so every reconstructed probability is physical and nonnegative.

The six constraints below have independent differentials, so the regular level-set theorem supplies local coordinates and integrates every tangent. Signature, time orientation, positive marginals, and full rank are open conditions, giving a two-sided physical curve for sufficiently small parameter values even when a joint probability is zero. $\square$

Proposition 6.2 (Smooth incidence manifold). *Restricted to the open genuine residual stratum—in particular, where g has Lorentz signature, all strict time-orientation inequalities hold, and P is invertible—the normalized incidence space is a smooth real 14-dimensional manifold.*

Proof.

$$\nabla_P F_j = 2QPr_j r_j^T.$$

If a linear combination vanishes, invertibility of QP gives

$$\sum_j \mu_j r_j r_j^T = \text{diag}(\mu_1, \dots, \mu_4) + \mu_5 r_5 r_5^T = 0.$$

Off-diagonal entries first give $\mu_5 = 0$, then diagonal entries give all other coefficients zero. Normalization is independent because the radial variation $\delta P = P$ annihilates every dF_j but changes $u^T P u$. There are $16 + 4 - 5 - 1 = 14$ dimensions. $\square$

Remark 6.3. The proposition is local to the open stratum used by the residual closure; singular normalized incidence points outside that stratum are not asserted to be smooth. The projected behavior image has dimension at most 14; no claim of exact projected dimension is needed for the closure theorem.

7 Stationarity and strictly positive multipliers

Every entry of the complete residual behavior is $r_i^T P r_j$. Thus, after collecting the original Bell coefficients into numbers γ_{ij} , its restriction to the incidence coordinates is the linear objective

$$L_C(P) := \sum_{i,j} \gamma_{ij} r_i^T P r_j = \text{Tr}(C^T P), \quad C := \sum_{i,j} \gamma_{ij} r_i r_j^T.$$

At a stationary point, the six constraint gradients are linearly independent. Thus there are unique multipliers $\alpha, \lambda_1, \dots, \lambda_5$, with sign convention

$$C - \alpha u u^T = -2QP\Lambda, \quad \Lambda = \sum_{j=1}^5 \lambda_j r_j r_j^T. \quad (5)$$

Stationarity in the four metric variables says $\lambda \in \ker \mathcal{D}^T$, where $\mathcal{D}$ is defined in section 8.

Lemma 7.1 (Finite POVM duality). *For*

$$\max_{N_j \geq 0, \sum_j N_j = \mathbb{I}} \sum_j \text{Tr}(K_j N_j),$$

there is a Hermitian Γ such that

$$\Gamma \geq K_j, \quad (\Gamma - K_j)N_j = 0.$$

Proof. Weak duality is immediate. Strong duality follows by separating the closed finite-dimensional cone

$$\left\{ \left(\sum_j X_j, \sum_j \text{Tr}(K_j X_j) - t \right) : X_j \geq 0, t \geq 0 \right\}$$

from $(\mathbb{I}, V + \varepsilon)$, where V is the primal optimum, and taking $\varepsilon \downarrow 0$. Equality of primal and dual values gives

$$\sum_j \text{Tr}[(\Gamma - K_j)N_j] = 0.$$

Every summand is nonnegative, hence zero; for positive matrices, $\text{Tr}(AB) = 0$ implies $AB = 0$. Full details are included in appendix E. $\square$

Proposition 7.2 (Strict multiplier positivity). *At a residual global Bell maximizer whose value is strictly above the PVM support value,*

$$\lambda_j > 0 \quad (j = 1, \dots, 5), \quad \Lambda > 0.$$

Proof. For each Bob input y , hold the state and all other measurements fixed and collect the terms of L_C depending on that input as

$$\sum_{j \in B_y} \text{Tr}(K_{j|y} N_{j|y}).$$

Within one input abbreviate $K_{j|y}$ and $N_{j|y}$ by K_j, N_j . Apply lemma 7.1 separately to Bob's binary and ternary inputs, with distinct normalization operators Γ_y . A nonzero rank-one qubit effect satisfies

$$\Gamma_y - K_j = s_j \text{adj}(N_j), \quad s_j \geq 0,$$

and $\text{adj}(N_j)N_j = 0$.

Under full-rank steering,

$$F_j = 4|\det C|^2 \det N_j.$$

Pulling (5) back to variations of the effects on input y gives

$$K_j - \Gamma_y + 4|\det C|^2 \lambda_j \text{adj}(N_j) = 0.$$

The normalization operator here is the dual operator: if two candidates satisfy complementary slackness, their difference annihilates the range of every effect in that input, and those ranges span $\mathbb{C}^2$ because the effects sum to $\mathbb{I}$. Therefore

$$4|\det C|^2 \lambda_j = s_j \geq 0.$$

If $s_j = 0$, then $K_j = \Gamma_y$. Replacing that entire input by the deterministic PVM $N_j = \mathbb{I}$, $N_k = 0$ preserves the score. That party now has only one nontrivial input. Every corresponding nonsignaling behavior is local: use the output of the nontrivial input as a hidden variable and sample the other party's potential outputs conditionally. Deterministic local behaviors use only identity and zero projectors. A strict value above the PVM support therefore forces every s_j , and hence every λ_j , to be positive. Since $r_1, \dots, r_4$ form a basis, $\Lambda > 0$. $\square$

8 Exact second variation

A tangent metric variation has the form

$$H(A, B, C, D) = \begin{pmatrix} 0 & 0 & A & B \\ 0 & 0 & C & D \\ A & C & 0 & A + B + C + D \\ B & D & A + B + C + D & 0 \end{pmatrix}.$$

Define

$$\Phi(y) = \begin{pmatrix} y_2(y_0 + y_3) \\ y_3(y_0 + y_2) \\ y_2(y_1 + y_3) \\ y_3(y_1 + y_2) \end{pmatrix}, \quad y_j = QPr_j,$$

and

$$\mathcal{D} : H \mapsto (y_1^T H y_1, \dots, y_5^T H y_5).$$

Then

$$\text{rank } \mathcal{D} = \dim \text{span}\{\Phi(y_1), \dots, \Phi(y_5)\}.$$

Let $K = \ker \mathcal{D}^T$, $k = \dim K = 5 - \text{rank } \mathcal{D}$. Metric stationarity says $\lambda \in K$.

For a tangent $(\delta P, H)$, set

$$S = P^T Q P, \quad \delta P = P W + H Q P.$$

The first-order equations are solvable for H exactly when

$$\text{Tr}(S W \Lambda_\mu) = 0 \quad (\mu \in K), \quad \Lambda_\mu = \sum_j \mu_j r_j r_j^T.$$

The compatible space

$$\mathcal{H}_K = \{W : \text{Tr}(S W \Lambda_\mu) = 0 \ \forall \mu \in K\}$$

has dimension $16 - k$.

Proposition 8.1 (Weighted second form).

$$\sum_j \lambda_j D^2 F_j = 2q(W), \quad q(W) = \text{Tr}(S W \Lambda W^T),$$

and

$$\text{inertia}(q) = (4, 12).$$

Every positive- q compatible tangent can be adjusted to satisfy normalization without changing q .

Proof. Differentiating g^{-1} twice and completing the square after $\delta P = P W + H Q P$ gives

$$D^2 F_j = 2(P W r_j)^T Q (P W r_j),$$

and summing against λ_j yields the trace formula. The exact expansion is recorded in appendix F and independently checked by the verifier.

The matrix S has signature $(1, 3)$, while $\Lambda > 0$. Invertible congruences therefore reduce q to

$$q_0(Z) = \sum_{j=1}^4 Z_{1j}^2 - \sum_{i=2}^4 \sum_{j=1}^4 Z_{ij}^2,$$

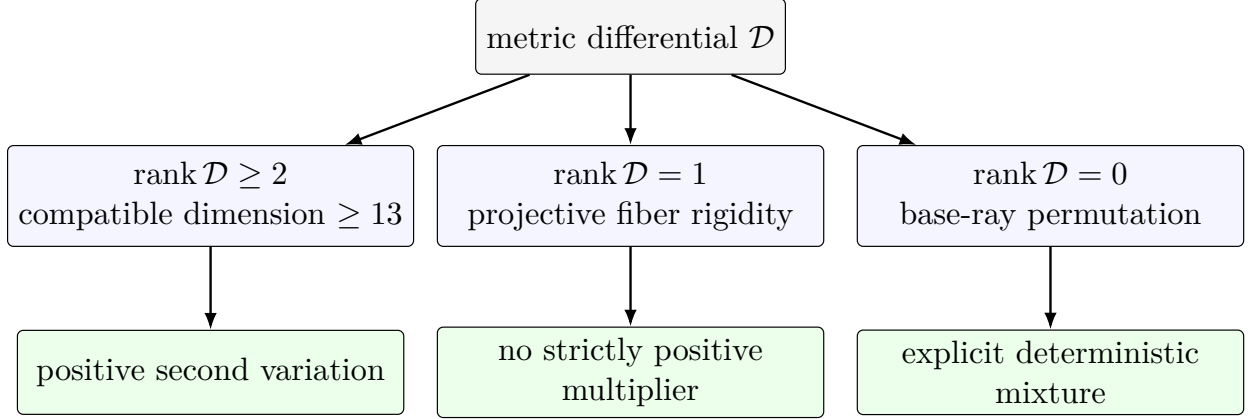
giving inertia $(4, 12)$.

The identity matrix belongs to $\mathcal{H}_K$, $q(\mathbb{I}) = 0$, and the polar cross term $\text{Tr}(S W \Lambda)$ vanishes because $\lambda \in K$. Thus $q(W + t\mathbb{I}) = q(W)$. Adding $t\mathbb{I}$ changes the normalization derivative by t , so choose t to impose $u^T \delta P u = 0$. $\square$

With the convention (5), a normalized physical curve has

$$\frac{d^2}{dt^2} L_C(P(t)) \Big|_{t=0} = 2q(W).$$

Hence local maximality requires $q \leq 0$ on every normalized tangent.



Every rank contradicts a strict non-PVM Bell maximum.

Figure 3: The exhaustive rank trichotomy for the metric differential.

9 Rank trichotomy and universal closure

9.1 Rank at least two

If $\text{rank } \mathcal{D} \geq 2$, then $k \leq 3$ and

$$\dim \mathcal{H}_K = 16 - k \geq 13.$$

A subspace on which a form of inertia $(4, 12)$ is nonpositive has dimension at most 12: projection to its twelve negative coordinates is injective. Therefore $\mathcal{H}_K$ contains W with $q(W) > 0$. Normalize it using proposition 8.1 and integrate it using theorem 6.1. The resulting physical curve has positive Bell second derivative, contradicting local maximality.

9.2 Rank one

The coefficient form of the metric differential is the quadratic map

$$\Phi(y) = (y_2(y_0 + y_3), y_3(y_0 + y_2), y_2(y_1 + y_3), y_3(y_1 + y_2))^T.$$

Proposition 9.1 (Projective fiber theorem). *On the strict g -null quadric, the base locus of Φ is exactly the five coefficient rays $[r_1], \dots, [r_5]$. Away from the base locus, Φ is projectively injective, including on the exceptional divisors*

$$y_2 = 0, \quad y_3 = 0, \quad y_0 = y_1, \quad y_2 = y_3.$$

Proof. The generic rational inverse and the exact exceptional-divisor calculations are given in appendix G. Each resultant and factorization is reproduced by the exact verifier. The strict metric inequalities exclude extra null roots without excluding any genuine residual point. $\square$

If $\text{rank } \mathcal{D} = 1$, all nonzero $\Phi(y_j)$ are projectively equal. The five $y_j = QPr_j$ are distinct rays because QP is invertible. Proposition 9.1 therefore permits at most one nonzero row of $\mathcal{D}$. Consequently

$$\ker \mathcal{D}^T = \{\mu : \mu_{j_0} = 0\}$$

for some j_0 , which contains no strictly positive vector. This contradicts $\lambda_j > 0$.

9.3 Rank zero

If $\text{rank } \mathcal{D} = 0$, every y_j belongs to the five-ray base locus. Invertibility gives

$$y_j = t_j r_{\pi(j)}, \quad t_j > 0,$$

for a permutation π . The unique circuit forces π to preserve the binary and ternary partitions and all t_j to be equal. Normalization fixes the common scale, so after relabeling

$$QP = \mathbb{I}, \quad P = g.$$

Proposition 9.2 (Rank-zero deterministic simulation). *Every rank-zero probability table $p_{ij} = r_i^T g r_j$ is a convex combination of deterministic local behaviors, and hence belongs to $\overline{\mathcal{Q}}_2^{\text{PVM}}$.*

Proof. Let c_{ij} be the ternary–ternary block, with $c_{ii} = 0$, and let a_i, b_i be the two binary–ternary rows. They obey

$$\sum_{j \neq i} c_{ij} = a_i + b_i, \quad \sum_i a_i = \sum_i b_i = \frac{1}{2}.$$

Appendix H constructs a matrix q_{ij} satisfying

$$0 \leq q_{ij} \leq c_{ij}, \quad \sum_j q_{ij} = b_i, \quad \sum_i q_{ij} = a_j.$$

For each ordered ternary pair (i, j) , assign weight q_{ij} to the deterministic binary outputs $(1, 2)$, and weight $c_{ij} - q_{ij}$ to $(2, 1)$, while fixing ternary outputs to (i, j) . Summing these branches reproduces the ternary block, both mixed blocks, and the binary block simultaneously. Identity and zero projectors implement every deterministic branch. $\square$

9.4 Equality

Theorem 9.3 (Universal two-input equality). *For every finite input-dependent output architecture $(A_0, A_1; B_0, B_1)$,*

$$\boxed{\overline{\mathcal{Q}}_2^{\text{POVM}}(\mathbf{A}, \mathbf{B}) = \overline{\mathcal{Q}}_2^{\text{PVM}}(\mathbf{A}, \mathbf{B}).}$$

Proof. The PVM-to-POVM inclusion is immediate. Suppose the reverse inclusion failed. The nearest-point argument in lemma 2.3 would give a Bell functional whose global POVM maximum strictly exceeds its PVM maximum. Choose an extreme point of its exposed maximizing face. By theorem 4.3, a maximizing component lies in the genuine residual architecture. Every local measurement is therefore globally optimal with the other variables fixed, so proposition 7.2 applies.

The metric differential has rank 0, 1, 2, 3, or 4. Rank at least two has an uphill physical direction; rank one cannot carry the required positive multiplier; rank zero is PVM-simulable by proposition 9.2. Every case contradicts a strict value above the PVM support function. $\square$

Corollary 9.4 (Minimum input architecture). *Up to exchanging the parties, 3×2 is the minimum input architecture for a strict fixed-qubit POVM-over-PVM Bell separation.*

Proof. With one input on either party every nonsignaling behavior is local, hence a deterministic PVM behavior maximizes every Bell functional. Two inputs on both parties are excluded by theorem 9.3. Theorem 3.1 supplies an exact 3×2 separation. $\square$

10 Operational consequences and limitations

For every Bell functional with two inputs per party, optimizing over fixed-qubit PVMs already provides the fixed-qubit POVM bound. Thus a value above that bound excludes all local-qubit realizations, not merely projective ones.

Likewise, no qubit behavior with two inputs per party can certify that a nonprojective local measurement was indispensable: after shared-randomness convexification, a PVM realization always exists. This does not identify the state or measurements in a particular laboratory realization, and it does not assert raw-set equality without shared randomness.

We do not claim the exact global POVM or PVM optimum of (1). We also do not infer a private-randomness entropy bound: such a statement requires an adversarial extension of the convex decomposition, not only equality of observed behavior sets.

11 Discussion

The threshold result is special to local dimension two. Natural open questions include minimum-input laws in dimension three and higher, multipartite scenarios, equality without convexification, constructive simulation algorithms for generic two-input behaviors, instruments and sequential measurements, and experimentally robust exact 3×2 witnesses. Determining the exact two-qubit POVM and PVM optima of (1) is also open but unnecessary for the separation.

The proof illustrates a distinction between operator-level and behavioral generality. Ternary qubit POVMs are genuine generalized measurements, yet two inputs do not provide enough Bell structure to make them indispensable after the state and shared-randomness decomposition may change. A third input can lock the state and steering geometry while reserving another measurement for a discrimination advantage.

Data, code, and verification

The accompanying repository contains exact Bell coefficients and strategies, the separation verifier, the exceptional-fiber identities, the closure verifier, and an independent rank-zero simulator. One offline command runs all exact checks. The code verifies finite algebraic identities and explicit constructions; the convexity, extremality, physical-reconstruction, and dimension arguments remain human proofs in this manuscript.

AI-assisted research and verification

Generative-AI systems were used extensively during mathematical exploration, symbolic derivation, adversarial proof analysis, software development, and preparation of the manuscript in July 2026. The final arguments and computational artifacts were assembled under the responsibility of the human author, who takes full responsibility for all claims, proofs, citations, and released materials. Generative-AI systems are not listed as authors. The included exact verifiers were implemented separately from several discovery calculations, but no claim of independent expert human verification is made.

Competing interests and funding

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A Exact algebra for the 3×2 separation

We record the identities that make the projective bound independent of numerical optimization. For a pair involving auxiliary label 2, let

$$A = \frac{3}{5}D\Pi(\mathbf{n})D, \quad B = \frac{4}{5}D\Pi(\mathbf{m})D,$$

where $x = n_z$, $y = m_z$, $r = \mathbf{n} \cdot \mathbf{m}$, and

$$D = \text{diag}\left(\sqrt{\frac{1+\eta}{2}}, \sqrt{\frac{1-\eta}{2}}\right).$$

Direct expansion, using $\|\mathbf{n}\| = \|\mathbf{m}\| = 1$, gives

$$\det(A - B) = -\frac{3}{50}(1 - \eta^2)(1 - r) \leq 0.$$

Thus $A - B$ has eigenvalues of opposite signs, including a possible zero boundary eigenvalue, and the optimized complementary-projector score is

$$F_\eta = \frac{\text{Tr } A + \text{Tr } B + \|A - B\|_1}{2}.$$

Its discriminant yields

$$20F_\eta = 7 + \eta(3x + 4y) + \sqrt{[-1 + \eta(3x - 4y)]^2 + 24(1 - \eta^2)(1 - r)}.$$

At $\eta = 0$,

$$F_0(r) = \frac{7}{20} + \frac{1}{20}\sqrt{25 - 24r}.$$

Set

$$q = \sqrt{25 - 24r} \geq 1, \quad H = 8 - 3x - 4y \geq 1.$$

To prove $F_\eta \leq F_0(r) + 2\eta/5$, it is enough to prove

$$\sqrt{[-1 + \eta(3x - 4y)]^2 + 24(1 - \eta^2)(1 - r)} \leq q + \eta H.$$

The right side is at least one, so squaring is valid. Using $q^2 = 25 - 24r$, the squared difference is exactly

$$2\eta\{\eta[8(1 - y)(4 - 3x) + 12(1 - r)] + (q - 1)(8 - 3x - 4y) + 8(1 - y)\}.$$

Every factor is nonnegative on

$$0 \leq \eta \leq 1, \quad -1 \leq x, y, r \leq 1.$$

No omitted compatibility condition among x, y, r can invalidate the estimate because it holds on the larger box.

For $u = |\mathbf{b}_0 \cdot \mathbf{b}_1|$, both relevant pairs satisfy $r \geq -u$. Moreover,

$$(5 + 8u)^2 - (25 + 24u) = 8u(7 + 8u) \geq 0,$$

so

$$F_0(r) \leq \frac{7}{20} + \frac{1}{20}\sqrt{25+24u} \leq \frac{3}{5} + \frac{2u}{5}.$$

For completeness, the scalar inequalities in the CHSH-deficit estimate have nonnegative squared differences

$$\begin{aligned} \left(1 - \frac{u^2}{2}\right)^2 - (1 - u^2) &= \frac{u^4}{4}, \\ \left(2 - \frac{u^2}{4}\right)^2 - (4 - u^2) &= \frac{u^4}{16}. \end{aligned}$$

Their right sides are nonnegative on $0 \leq u \leq 1$, so no sign reversal is introduced by squaring.

B The strengthened algebraic strategy

Put

$$q = \sqrt{7813}, \quad \eta = \frac{1}{q}, \quad s = \sqrt{1 - \eta^2} = \frac{6\sqrt{217}}{q}.$$

Use

$$|\psi_\eta\rangle = \sqrt{\frac{1+\eta}{2}}|00\rangle + \sqrt{\frac{1-\eta}{2}}|11\rangle,$$

Bob observables $B_0 = X, B_1 = Z$, and Alice observables

$$A_0 = \frac{6\sqrt{217}X + qZ}{125}, \quad A_1 = \frac{6\sqrt{217}X - qZ}{125}.$$

They square to the identity because

$$36 \cdot 217 + 7813 = 125^2.$$

The CHSH value is $S = 250/q$.

Set

$$k = \frac{3}{5}\sqrt{\frac{q-1}{q+1}}$$

and

$$M_0 = \frac{1}{2} \begin{pmatrix} k^2 & k \\ k & 1 \end{pmatrix}, \quad M_1 = \frac{1}{2} \begin{pmatrix} k^2 & -k \\ -k & 1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 - k^2 & 0 \\ 0 & 0 \end{pmatrix}.$$

These are positive rank-one effects and sum to the identity. The diagonal dual operator

$$\Gamma = \text{diag}\left(\frac{2}{5}(1+\eta), \frac{6}{25}(1-\eta)\right)$$

dominates all three weighted steered score operators and annihilates the corresponding effect ranges. Hence

$$T = \text{Tr } \Gamma = \frac{16+4\eta}{25}, \quad 10S + T = \frac{16+8q}{25}.$$

Within this one-parameter family,

$$f(\eta) = 20\sqrt{2-\eta^2} + \frac{16+4\eta}{25}$$

has

$$f'(\eta) = -\frac{20\eta}{\sqrt{2-\eta^2}} + \frac{4}{25}, \quad f''(\eta) = -\frac{40}{(2-\eta^2)^{3/2}} < 0.$$

Its unique critical point is $\eta = 1/\sqrt{7813}$. This is not a global POVM optimality claim.

Finally,

$$(6 + 80\sqrt{7813})^2 - 2 \cdot 5003^2 = -56782 + 960\sqrt{7813},$$

and

$$960^2 \cdot 7813 - 56782^2 = 3\,976\,265\,276 > 0.$$

Both sides being compared are positive, these two exact square comparisons prove $L_1 > U$.

C Degenerate cases in the one-binary-party theorem

If B_0, B_1 are nondegenerate and have linearly independent Bloch axes, Φ_B maps the positive cone onto a three-dimensional Lorentz cone. An extreme ray of the image lies on the boundary ellipse. Its minimum-norm Bloch lift in the plane of the two axes has norm one; a perpendicular component would exceed the Bloch ball. Thus the positive lift in $\text{span}_{\mathbb{R}}\{\mathbb{I}, B_0, B_1\}$ is unique and rank one.

If the Bloch axes are parallel or antiparallel, the image is the cone over an interval and has two extreme rays. Every positive relation reduces to one-versus-one cancellations or a two-versus-two relation with repeated zero-padded outcomes. The same purification construction applies.

If one observable is scalar and the other nondegenerate, the image is again two-dimensional; the scalar input is deterministic and does not add a coordinate. If both are scalar, Bob is deterministic on both inputs and the behavior is local. A rank-deficient circuit sum Ω forces all lifted rays to be collinear and is realized by a product purification. Zero effects receive zero projectors. These cases also show that circuit weights remain nonnegative and sum to one at the boundary.

D Extremal qubit POVMs and the rank-square bound

Let M_a be the nonzero effects of a POVM. For each a , let $\mathcal{V}_a$ be the real vector space of Hermitian operators supported on $\text{ran } M_a$. It has dimension $\text{rank}(M_a)^2$. Consider the summation map

$$\bigoplus_a \mathcal{V}_a \longrightarrow \text{Herm}(2), \quad (H_a)_a \longmapsto \sum_a H_a.$$

If it has a nonzero kernel, then for sufficiently small $\varepsilon > 0$, $M_a \pm \varepsilon H_a \geq 0$, and the two perturbed tuples are distinct POVMs whose average is M . Extremality therefore requires injectivity, giving

$$\sum_a \text{rank}(M_a)^2 \leq \dim_{\mathbb{R}} \text{Herm}(2) = 4.$$

This necessary condition is all that is used in theorem 4.3. Together with linear independence in a three-dimensional span, it leaves exactly three rank-one effects.

E Finite-dimensional POVM duality

Let

$$V = \max_{\substack{N_j \geq 0 \\ \sum_j N_j = \mathbb{I}}} \sum_j \text{Tr}(K_j N_j).$$

Weak duality says that every $\Gamma \geq K_j$ obeys

$$\sum_j \text{Tr}(K_j N_j) \leq \text{Tr } \Gamma.$$

For the reverse direction, consider the closed cone

$$\mathcal{C} = \left\{ \left(\sum_j X_j, \sum_j \text{Tr}(K_j X_j) - t \right) : X_j \geq 0, t \geq 0 \right\} \subset \text{Herm}(2) \times \mathbb{R}.$$

Closedness follows because convergence of $\sum_j X_j$ bounds every $0 \leq X_j \leq \sum_k X_k$.

The point $(\mathbb{I}, V + \varepsilon)$ is outside $\mathcal{C}$. Nearest-point separation provides a nonzero functional

$$\ell(A, s) = \text{Tr}(\Gamma_\varepsilon A) + \gamma s$$

nonnegative on $\mathcal{C}$ and negative at that point. Since $(0, -t) \in \mathcal{C}$, $\gamma \leq 0$. If $\gamma = 0$, positivity on all $(X, \text{Tr}(K_j X))$ would give $\Gamma_\varepsilon \geq 0$, contradicting $\text{Tr } \Gamma_\varepsilon < 0$. Normalize $\gamma = -1$. Taking one nonzero $X_j = X$ gives $\Gamma_\varepsilon \geq K_j$, while separation and weak duality give

$$V \leq \text{Tr } \Gamma_\varepsilon < V + \varepsilon.$$

A bounded convergent subsequence as $\varepsilon \downarrow 0$ yields dual attainment with $\text{Tr } \Gamma = V$.

At a primal optimum,

$$0 = \text{Tr } \Gamma - \sum_j \text{Tr}(K_j N_j) = \sum_j \text{Tr}[(\Gamma - K_j) N_j].$$

Each term is nonnegative, hence zero. For $A, B \geq 0$, $\text{Tr}(AB) = 0$ implies $A^{1/2} B A^{1/2} = 0$, and therefore $AB = 0$. This proves complementary slackness.

F Square completion for the incidence Hessian

Let $x_j = P r_j$, $Q = g^{-1}$, and vary

$$P(t) = P + t\delta P, \quad g(t) = g + tH$$

to first order. The inverse expansion is

$$(g + tH)^{-1} = Q - tQHQ + t^2QHQHQ + O(t^3).$$

Substitution in $F_j = (P r_j)^T Q (P r_j)$ gives

$$\begin{aligned} \frac{1}{2} D^2 F_j &= (\delta P r_j)^T Q (\delta P r_j) - 2(\delta P r_j)^T Q H Q (P r_j) \\ &\quad + (P r_j)^T Q H Q H Q (P r_j). \end{aligned}$$

Writing

$$\delta P = P W + H Q P$$

turns the right side into

$$(P W r_j)^T Q (P W r_j).$$

Therefore

$$\sum_j \lambda_j D^2 F_j = 2 \sum_j \lambda_j r_j^T W^T P^T Q P W r_j = 2 \text{Tr}(S W \Lambda W^T).$$

This expansion also fixes the sign convention used at a Bell maximum.

G Projective fibers of the quadratic null-ray map

Let

$$\mathcal{N}_g = \{[x] \in \mathbb{RP}^3 : x^T g x = 0\}$$

for a strict metric (3), and write $\Phi(x) = (A, B, C, D)$.

G.1 Base locus and generic inverse

If $x_2 = x_3 = 0$, nullness gives $x_0 x_1 = 0$, hence r_1, r_2 . If exactly one of x_2, x_3 vanishes, the remaining equations give r_3 or r_4 . If $x_2 x_3 \neq 0$, all four components of Φ vanish only when

$$x_0 = x_1 = -x_2 = -x_3,$$

the ray r_5 . Thus the base locus consists exactly of the five coefficient rays.

The identities

$$\begin{aligned} A - B &= x_0(x_2 - x_3), & C - D &= x_1(x_2 - x_3), \\ A - C &= x_2(x_0 - x_1), & B - D &= x_3(x_0 - x_1), \\ AD - BC &= x_2 x_3(x_0 - x_1)(x_2 - x_3) \end{aligned}$$

give

$$\begin{pmatrix} (A - B)(A - C)(B - D) \\ (C - D)(A - C)(B - D) \\ (AD - BC)(A - C) \\ (AD - BC)(B - D) \end{pmatrix} = \omega(x)x,$$

where

$$\omega(x) = x_2 x_3(x_2 - x_3)(x_0 - x_1)^2.$$

Hence $[x]$ is recovered from $[\Phi(x)]$ whenever $\omega(x) \neq 0$.

G.2 Exceptional divisor $x_2 = 0$

The target is $[0 : B : 0 : D]$. On the direct branch, scale $z_3 = 1$ and write $(z_0, z_1) = t(B, D)$. Nullness is

$$t^2 BD + 2t(bB + dD) = 0.$$

The root $t = 0$ is r_4 , leaving at most one nonbase root. On the cross branch $z_2 \neq 0$, target zeros force $z_0 = z_1 = -z_3$ and a nonzero target requires $B = D$. With $z_3 = 1, z_2 = t$,

$$z^T g z = (1 - 2(b + d))(1 - t).$$

Strictness $b + d < 1/2$ gives only $t = 1$, the base ray $-r_5$.

G.3 Exceptional divisor $x_3 = 0$

The symmetric direct equation is

$$t^2 AC + 2t(aA + cC) = 0,$$

and the cross equation is

$$(1 - 2(a + c))(1 - t) = 0.$$

Again there is at most one nonbase preimage.

G.4 Exceptional divisor $x_0 = x_1$

If $x_0 = x_1 = 0$, nullness reduces to $2ex_2x_3 = 0$. Since $e > 0$, the point is r_3 or r_4 , hence in the base locus. A nonbase point may therefore be scaled to $x_0 = x_1 = 1$. Put $p = x_2, q = x_3$. The null and target-ratio equations are

$$N = 1 + 2(a + c)p + 2(b + d)q + 2epq = 0,$$

$$R = Bp(1 + q) - Aq(1 + p) = 0.$$

Their exact resultant in p is

$$\text{Res}_p(N, R) = -(q + 1) [B + q(A(2a + 2c - 1) + 2B(b + d))].$$

At $q = -1$, strictness gives $p = -1$, the base ray. The second factor is not identically zero: if $B \neq 0$ its constant term is nonzero; if $B = 0$, then $A \neq 0$ and $2(a + c) - 1 \neq 0$. Once q is fixed, N determines at most one p . The only apparent failure would require

$$2(a + c)(b + d) = a + b + c + d - \frac{1}{2},$$

equivalent to

$$(2(a + c) - 1)(2(b + d) - 1) = 0,$$

which strictness excludes.

G.5 Exceptional divisor $x_2 = x_3$

If $x_2 = x_3 = 0$, nullness gives $x_0x_1 = 0$, hence the base rays r_1, r_2 . For a nonbase point scale $x_2 = x_3 = 1$ and put $p = x_0, q = x_1$. The equations are

$$N = pq + 2(a + b)p + 2(c + d)q + 2e = 0,$$

$$R = C(p + 1) - A(q + 1) = 0.$$

The exact resultant is

$$\text{Res}_p(N, R) = -(q + 1) [Aq + 2A(a + b) + 2C(c + d) - C].$$

At $q = -1$, strictness gives $p = -1$, the base ray. Strictness also excludes an identically vanishing second factor or an undetermined null equation.

For a possible cross-branch preimage with $z_2 \neq z_3$, equal target coordinates give $z_0 = z_1 = 0$; strict $e > 0$ and nullness then force $z_2z_3 = 0$, again a base ray. These cases exhaust $\omega = 0$, proving projective injectivity away from the base locus.

H Bounded transportation and the rank-zero simulation

For three ternary labels, let $c_{ij} = c_{ji} \geq 0$, $c_{ii} = 0$, and assume

$$\sum_{j \neq i} c_{ij} = a_i + b_i, \quad \sum_i a_i = \sum_i b_i = \frac{1}{2}.$$

Seek q_{ij} with row sums b_i , column sums a_j , and $0 \leq q_{ij} \leq c_{ij}$. Put

$$q_{ij} = \frac{c_{ij}}{2} + f_{ij}, \quad f_{ij} = -f_{ji},$$

$$d_i = \frac{b_i - a_i}{2}, \quad d_1 + d_2 + d_3 = 0.$$

Set

$$f_{12} = t, \quad f_{13} = d_1 - t, \quad f_{23} = d_2 + t.$$

The row and column equations are automatic. Capacity is equivalent to

$$t \in I_{12} \cap I_{13} \cap I_{23},$$

where

$$\begin{aligned} I_{12} &= [-c_{12}/2, c_{12}/2], \\ I_{13} &= [d_1 - c_{13}/2, d_1 + c_{13}/2], \\ I_{23} &= [-d_2 - c_{23}/2, -d_2 + c_{23}/2]. \end{aligned}$$

The inequalities

$$|d_i| \leq \frac{a_i + b_i}{2} = \frac{c_{ij} + c_{ik}}{2}$$

show that every pair of intervals intersects, including zero-capacity endpoints. Three closed intervals on the real line with pairwise intersections have a common point: the largest left endpoint does not exceed the smallest right endpoint. Any common t gives the required bounded transportation matrix.

The deterministic branch construction in proposition 9.2 then reproduces all four input blocks simultaneously. The supplied rational simulator independently checks the construction on an interior symmetric instance; generality follows from the interval proof, not from that one executable example.

I Verification boundary

The exact verifier checks:

1. dense and sparse Bell coefficients and both explicit strategies;
2. POVM normalization, rank-one factorizations, exact Bell values, and discrimination dual certificates;
3. the determinant, discriminant, robust-comparison, deficit, and complete-square identities in the projective bound;
4. the Lorentz metric identities, quadratic map, generic inverse, all exceptional resultants and factorizations;
5. an exact pure-state conformal-Lorentz instance and the weighted Hessian square completion;
6. independence witnesses for the five incidence constraints;
7. bounded-flow identities and an independent rank-zero simulation.

The scripts do not replace the human arguments establishing compactness, extremal reduction, cone circuits, physical completeness, multiplier signs, dimension bounds, or exhaustion of cases. Some symbolic checks are exact regression instances rather than formal proofs of general matrix identities; the corresponding general identities are derived above.

References

- [1] Nicolas Brunner, Daniel Cavalcanti, Stefano Pironio, Valerio Scarani, and Stephanie Wehner. Bell nonlocality. *Reviews of Modern Physics*, 86(2):419–478, 2014. doi: 10.1103/RevModPhys.86.419. URL <https://doi.org/10.1103/RevModPhys.86.419>.
- [2] Miguel Navascués, Adrien Feix, Mateus Araújo, and Tamás Vértesi. Characterizing finite-dimensional quantum behavior. *Physical Review A*, 92(4):042117, 2015. doi: 10.1103/PhysRevA.92.042117. URL <https://doi.org/10.1103/PhysRevA.92.042117>.
- [3] Tamás Vértesi and Erika Bene. Two-qubit Bell inequality for which positive operator-valued measurements are relevant. *Physical Review A*, 82(6):062115, 2010. doi: 10.1103/PhysRevA.82.062115. URL <https://doi.org/10.1103/PhysRevA.82.062115>.
- [4] Johanna F. Barra, Esteban S. Gómez, Gustavo Cañas, Waldimar A. T. Nogueira, Leonardo Neves, and Gustavo Lima. Higher quantum bound for the Vértesi–Bene–Bell inequality and the role of positive operator-valued measures regarding its threshold detection efficiency. *Physical Review A*, 86(4):042114, 2012. doi: 10.1103/PhysRevA.86.042114. URL <https://doi.org/10.1103/PhysRevA.86.042114>.
- [5] Esteban S. Gómez, Santiago Gómez, Pablo González, Gustavo Cañas, Johanna F. Barra, Aldo Delgado, Guilherme B. Xavier, Adán Cabello, Matthias Kleinmann, Tamás Vértesi, and Gustavo Lima. Device-independent certification of a nonprojective qubit measurement. *Physical Review Letters*, 117(26):260401, 2016. doi: 10.1103/PhysRevLett.117.260401. URL <https://doi.org/10.1103/PhysRevLett.117.260401>.
- [6] Michał Oszmaniec, Leonardo Guerini, Peter Wittek, and Antonio Acín. Simulating positive-operator-valued measures with projective measurements. *Physical Review Letters*, 119(19):190501, 2017. doi: 10.1103/PhysRevLett.119.190501. URL <https://doi.org/10.1103/PhysRevLett.119.190501>.
- [7] Leonardo Guerini, Jessica Bavaresco, Marcelo Terra Cunha, and Antonio Acín. Operational framework for quantum measurement simulability. *Journal of Mathematical Physics*, 58(9):092102, 2017. doi: 10.1063/1.4994303. URL <https://doi.org/10.1063/1.4994303>.
- [8] Lluís Masanes. Extremal quantum correlations for N parties with two dichotomic observables per site, 2005. URL <https://arxiv.org/abs/quant-ph/0512100>.
- [9] Benjamin Toner and Frank Verstraete. Monogamy of Bell correlations and Tsirelson’s bound, 2006. URL <https://arxiv.org/abs/quant-ph/0611001>.
- [10] Sacha Schwarz, Bänz Bessire, André Stefanov, and Yeong-Cherng Liang. Bipartite Bell inequalities with three ternary-outcome measurements—from theory to experiments. *New Journal of Physics*, 18(3):035001, 2016. doi: 10.1088/1367-2630/18/3/035001. URL <https://doi.org/10.1088/1367-2630/18/3/035001>.
- [11] Sacha Schwarz, Bänz Bessire, André Stefanov, and Yeong-Cherng Liang. Corrigendum: Bipartite Bell inequalities with three ternary-outcome measurements—from theory to experiments. *New Journal of Physics*, 20(4):049502, 2018. doi: 10.1088/1367-2630/aabe5c. URL <https://doi.org/10.1088/1367-2630/aabe5c>. The correction is also incorporated into arXiv:1511.05253v3.
- [12] Giacomo Mauro D’Ariano, Paoloplacido Lo Presti, and Paolo Perinotti. Classical randomness in quantum measurements. *Journal of Physics A: Mathematical and General*, 38(26):5979–5991, 2005. doi: 10.1088/0305-4470/38/26/010. URL <https://doi.org/10.1088/0305-4470/38/26/010>.
- [13] Erkkä Haapasalo, Teiko Heinosaari, and Juha-Pekka Pellonpää. Quantum measurements on finite dimensional systems: Relabeling and mixing. *Quantum Information Processing*, 11(6):1751–1763, 2012. doi: 10.1007/s11128-011-0330-2. URL <https://doi.org/10.1007/s11128-011-0330-2>.